

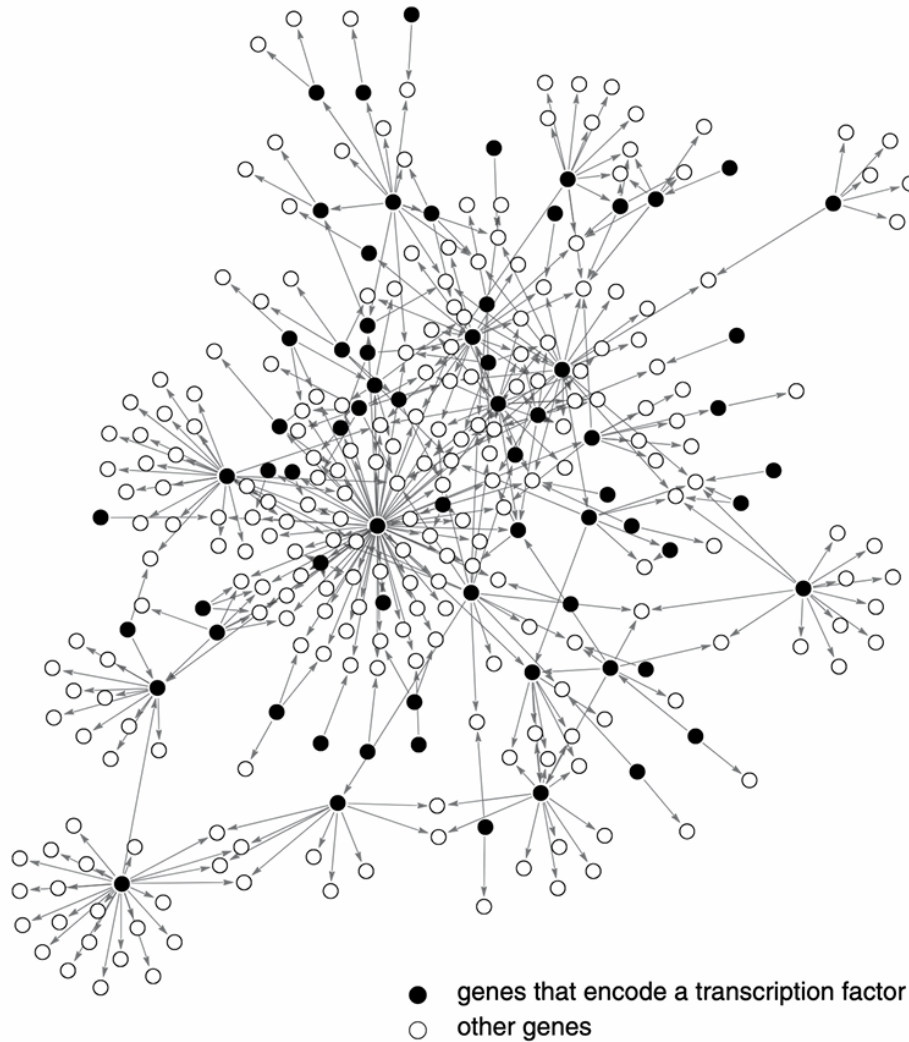
Introduction to Quantitative Biology
Lecture Notes: Gene Regulatory Network and
Response Time

from An Introduction to Systems Biology - Uri Alon

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1. From a complex gene regulatory network to a quantitative question

A transcriptional regulatory network contains many genes, transcription factors, and regulatory interactions. At the molecular level, every arrow can contain its own binding reactions, signaling steps, transcription, mRNA turnover, translation, and protein removal. Directly tracking all of these molecular details quickly becomes impractical.



Transcription network

*Figure 1. A transcriptional regulatory network contains many interacting regulatory edges. Black nodes denote genes encoding transcription factors. Source: Alon, **An Introduction to Systems Biology**, 2nd ed., Fig. 1.5.*

The central question for this lecture is therefore not how to reproduce every molecular event, but how to identify the slow variables that control the dynamics of the cellular response.

Question

A signal arrives at the cell. In the previous lecture, we estimated that making one functional protein molecule can take only tens of seconds, roughly

$$t_{\text{synthesis}} \sim 30 \text{ s.}$$

But the appearance of the first functional protein molecule is not yet the same as asking when the **protein concentration of the whole cell reaches its new steady state**.

How long does it take for a cellular response to approach steady state, and what determines this response time?

To answer this, we coarse-grain the gene regulatory network.

2. Modelling assumptions: what do we keep and what do we eliminate?

We make five modelling choices.

1. **Keep the topology of the regulatory network.** We preserve which regulator controls which target.
2. **Use transcription factors (TFs) to track the internal state of the cell.** TF activities provide a low-dimensional representation of the environmental and intracellular signals relevant for transcription.
3. **Focus on the hour-scale dynamics of protein concentrations.** Faster molecular processes are not followed explicitly unless they control this slow dynamics.
4. **Replace fast processes by input-output functions.** Promoter binding, signaling, and other fast regulatory chemistry are summarized by effective functions such as $f(x)$.
5. **Account for cell growth through dilution of intracellular concentrations.** Growth enters the protein-concentration dynamics as a removal term.

The key modelling move is to replace fast molecular details by effective terms, and then describe the slow protein-concentration dynamics with an ordinary differential equation.

3. Coarse-graining one regulatory edge

3.1 Notation

We use the following notation throughout the derivation:

- X : transcription factor; x denotes its concentration;
- D : free promoter / DNA binding site;

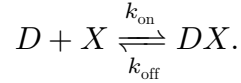
- DX : TF-bound promoter;
- $D_T = D + DX$: total promoter concentration;
- $k_{\text{on}}, k_{\text{off}}$: TF-DNA binding and unbinding rate constants;
- $K_d = k_{\text{off}}/k_{\text{on}}$: dissociation constant;
- $f(x)$: effective regulatory input-output function;
- m : mRNA concentration;
- Y : abundance (copy number) of protein Y in one cell;
- $[Y]$: concentration of protein Y ;
- y : shorthand for protein concentration, defined identically as

$$y \equiv [Y] \equiv \frac{Y}{V}$$

- V : cell volume;
- β : effective maximal protein-production rate per unit volume;
- γ_m : mRNA degradation rate;
- γ : protein degradation rate;
- μ : fractional cell-volume growth rate, $\mu \equiv \frac{1}{V} \frac{dV}{dt}$;
- $\alpha = \gamma + \mu$: total effective removal rate of protein concentration.

3.2 Timescale separation I: TF-DNA binding is fast

Consider TF binding to a promoter:



Mass-action kinetics gives

$$\frac{d[DX]}{dt} = k_{\text{on}}x[D] - k_{\text{off}}[DX].$$

TF binding and unbinding are much faster than the hour-scale changes in protein concentration. On the slow timescale of interest, we therefore approximate the binding reaction as being at quasi-steady state:

$$\frac{d[DX]}{dt} \approx 0.$$

Hence

$$k_{\text{on}}x[D] = k_{\text{off}}[DX],$$

and, with

$$K_d \equiv \frac{k_{\text{off}}}{k_{\text{on}}},$$

we obtain

$$[DX] = \frac{x[D]}{K_d}.$$

Using promoter conservation,

$$D_T = [D] + [DX],$$

we find the bound and unbound promoter fractions:

$$\boxed{\frac{[DX]}{D_T} = \frac{x}{K_d + x}}$$

and

$$\boxed{\frac{[D]}{D_T} = \frac{K_d}{K_d + x}}.$$

The microscopic binding reaction has therefore been replaced by an effective input-output relation.

If X is an activator, production can be proportional to the **bound** fraction. If X is a repressor, production can be proportional to the **unbound** fraction. We summarize either case by a dimensionless regulatory function $f(x)$.

3.3 Hill and MWC input functions

For cooperative regulation, a commonly used phenomenological input-output form is the **Hill function**. For example,

$$f_{\text{act}}(x) = \frac{x^n}{K^n + x^n},$$

whereas a repressive form is

$$f_{\text{rep}}(x) = \frac{K^n}{K^n + x^n}.$$

Here K sets the characteristic concentration scale and n controls the steepness of the transition. We will use these functions as compact input-output descriptions rather than derive their microscopic origin in detail.

A more mechanistic treatment of cooperative allosteric regulation is provided by the **Monod-Wyman-Changeux (MWC) model**.

Further reading: Alon, *An Introduction to Systems Biology*, Appendix A; Monod, Wyman & Changeux, *Journal of Molecular Biology* **12**, 88-118 (1965).

3.4 Timescale separation II: mRNA reaches quasi-steady state

We can write mRNA dynamics as

$$\frac{dm}{dt} = k_{\text{tx}}f(x) - \gamma_m m,$$

where k_{tx} is the maximal transcription rate.

If mRNA turnover is fast compared with the slower protein-concentration dynamics, then on the protein timescale

$$\frac{dm}{dt} \approx 0.$$

Therefore,

$$m \approx m_{\text{ss}} = \frac{k_{\text{tx}}}{\gamma_m} f(x).$$

If each mRNA produces protein at rate k_{tl} , then the protein production rate per unit volume is

$$k_{\text{tl}}m \approx \frac{k_{\text{tl}}k_{\text{tx}}}{\gamma_m} f(x).$$

Define the effective production constant

$$\beta \equiv \frac{k_{\text{tl}}k_{\text{tx}}}{\gamma_m}.$$

Then transcription and translation collapse into a single effective production term:

$\text{protein production rate} = \beta f(x).$

This is the second major coarse-graining step.

4. Cell growth produces dilution of protein concentration

Protein abundance and protein concentration are not the same quantity. We define

$y \equiv [Y] \equiv \frac{Y}{V}.$

The abundance of protein Y changes through production and active degradation:

$$\frac{dY}{dt} = \beta f(x)V - \gamma Y.$$

Differentiate the concentration directly:

$$\frac{dy}{dt} = \frac{d}{dt} \left(\frac{Y}{V} \right) = \frac{1}{V} \frac{dY}{dt} - \frac{Y}{V^2} \frac{dV}{dt}.$$

Substitute the protein-abundance equation:

$$\frac{dy}{dt} = \frac{\beta f(x)V - \gamma Y}{V} - \frac{Y}{V^2} \frac{dV}{dt}.$$

Using

$$y = \frac{Y}{V}$$

and defining the fractional volume-growth rate

$$\mu \equiv \frac{1}{V} \frac{dV}{dt},$$

we obtain

$$\frac{dy}{dt} = \beta f(x) - (\gamma + \mu)y.$$

Define

$$\alpha \equiv \gamma + \mu.$$

The entire regulatory edge therefore collapses to

$$\frac{dy}{dt} = \beta f(x) - \alpha y.$$

Comment: The term $-\gamma y$ represents active protein degradation. The term $-\mu y$ represents dilution caused by cell growth: the protein molecules are not chemically destroyed, but the same abundance is distributed over an increasing cell volume.

This is the key dynamical equation of the lecture: a complicated chain of molecular processes has become a simple **production-removal ODE** whose terms have clear physical meanings.

5. Rate plots and stability

For the rate-plot analysis of simple regulation, we consider the gene after a strong input has switched it to the ON state. In this regime, the regulatory input function reaches its maximal value, $f(x) = 1$, so the production term $\beta f(x)$ reduces to the constant production rate β .

Therefore,

$$r_{\text{prod}} = \beta,$$

whereas the removal rate increases linearly with protein concentration,

$$r_{\text{rem}} = \alpha y.$$

The intersection of the two curves satisfies

$$\beta = \alpha y_{\text{ss}},$$

so

$$y_{\text{ss}} = \frac{\beta}{\alpha}.$$

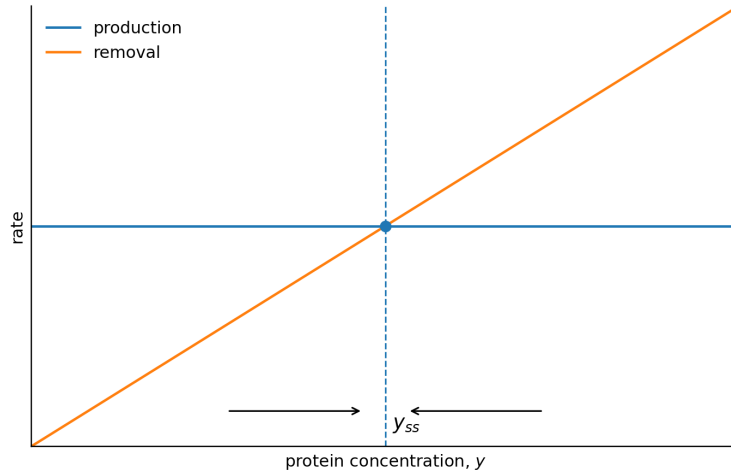


Figure 2. One-dimensional rate plot for simple gene-expression dynamics.

If $y < y_{\text{ss}}$, production exceeds removal and $dy/dt > 0$. If $y > y_{\text{ss}}$, removal exceeds production and $dy/dt < 0$. Both sides therefore flow back toward the same fixed point: the steady state is stable.

6. Response time: how fast does the concentration approach steady state?

For the same ON-state simple-regulation case used in the rate plot, the production rate is the constant β . The ODE is

$$\frac{dy}{dt} = \beta - \alpha y.$$

The new steady state is

$$y_{ss} = \frac{\beta}{\alpha}.$$

Rewrite the ODE as

$$\frac{dy}{dt} = -\alpha(y - y_{ss}).$$

Separate variables:

$$\frac{dy}{y - y_{ss}} = -\alpha dt.$$

Integrating from $t = 0$ to t ,

$$\int_{y_0}^{y(t)} \frac{dy'}{y' - y_{ss}} = -\alpha \int_0^t dt'.$$

Therefore,

$$\ln \left| \frac{y(t) - y_{ss}}{y_0 - y_{ss}} \right| = -\alpha t,$$

and

$$\boxed{y(t) = y_{ss} + (y_0 - y_{ss})e^{-\alpha t}.$$

Define the effective response time $T_{1/2}$ as the time required to move halfway from the initial level to the final steady state:

$$|y(T_{1/2}) - y_{ss}| = \frac{1}{2} |y_0 - y_{ss}|.$$

Hence

$$e^{-\alpha T_{1/2}} = \frac{1}{2},$$

so

$$T_{1/2} = \frac{\ln 2}{\alpha} = \frac{\ln 2}{\gamma + \mu}.$$

This is the central result.

The response time is controlled by protein removal: active degradation plus dilution by growth.

The promoter strength β determines the final steady-state concentration in the ON state,

$$y_{\text{ss}} = \frac{\beta}{\gamma + \mu},$$

but it cancels from the fractional response time:

$$T_{1/2} \text{ is independent of promoter strength in this simple model.}$$

Special case: a stable protein

For a stable protein with negligible active degradation,

$$\gamma \approx 0,$$

so

$$T_{1/2} = \frac{\ln 2}{\mu}.$$

If the fractional growth rate μ is approximately constant, the cell-volume doubling time is

$$T_{\text{dbl}} = \frac{\ln 2}{\mu}.$$

Therefore,

$$T_{1/2} = T_{\text{dbl}}$$

for a stable protein in this simple growing-cell model. Thus, when active degradation is negligible, growth dilution alone sets a response timescale of approximately one cell generation.

7. What removal process dominates in real cells?

The effective removal rate is the **sum of rates**,

$$\alpha = \gamma + \mu.$$

If two removal timescales are very different, the process with the **larger rate**, equivalently the **shorter characteristic time**, tends to dominate the response.

In bacteria, the degradation rates of both mRNA and protein are not fixed but regulated. Each is set by dedicated enzymes whose abundance and activity control the corresponding turnover rate.

The following are the rounded lecture-board values used in class. They should be interpreted as order-of-magnitude timescales rather than universal constants.

7.1 mRNA turnover

System	mRNA degradation timescale	Cell-cycle time
Bacteria	~ 3 min	~ 1 h
Yeast	~ 30 min	~ 3 h
Human cell line (HeLa)	~ 300 min	~ 1 day

For these lecture-scale values, mRNA degradation is faster than dilution, supporting the quasi-steady-state approximation used above.

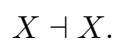
7.2 Protein turnover

System	Protein degradation timescale	Cell-cycle time
Bacteria	a few hours	~ 1 h
Yeast	~ 1 day	~ 3 h
HeLa	several days	~ 1 day

For many stable proteins, dilution can therefore be as fast as, or faster than, active protein degradation. This explains why the cell cycle can directly set the response time of protein concentration.

8. Negative autoregulation: two useful qualitative effects

Negative autoregulation (NAR) means that a transcription factor represses the promoter that produces itself:



We will not derive the NAR dynamics here. Instead, the two main effects can be understood graphically.

8.1 NAR can speed up the response

When X is initially low, self-repression is weak, so the promoter can produce X rapidly. As X accumulates, repression becomes stronger and acts as a brake near the desired steady state.

The rate-plot geometry makes this intuition precise. For a controlled comparison, take the removal rate and the final steady state to be the same in simple regulation and negative autoregulation. The NAR production curve is a decreasing function of X . For $X < X_{\text{st}}$, the vertical distance between production and removal is larger for NAR, so

$$\left(\frac{dX}{dt}\right)_{\text{NAR}} > \left(\frac{dX}{dt}\right)_{\text{simple}}.$$

Thus the system moves toward the same fixed point more rapidly.

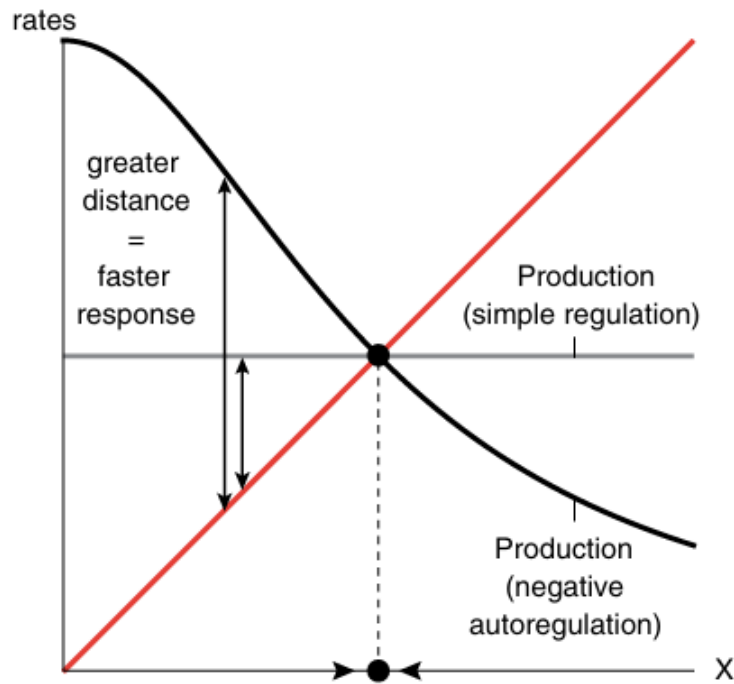


Figure 3. Rate-plot geometry of negative autoregulation. The larger gap between production and removal for $X < X_{\text{st}}$ gives a faster response. Source: Alon, *An Introduction to Systems Biology*, 2nd ed., Fig. 2.9.

8.2 NAR can reduce cell-to-cell variation

Different cells can have somewhat different production or removal rates. With simple regulation, these differences can move the fixed point over a broad range of protein concentrations. A decreasing NAR production curve makes the intersection points cluster more tightly, buffering those parameter differences.

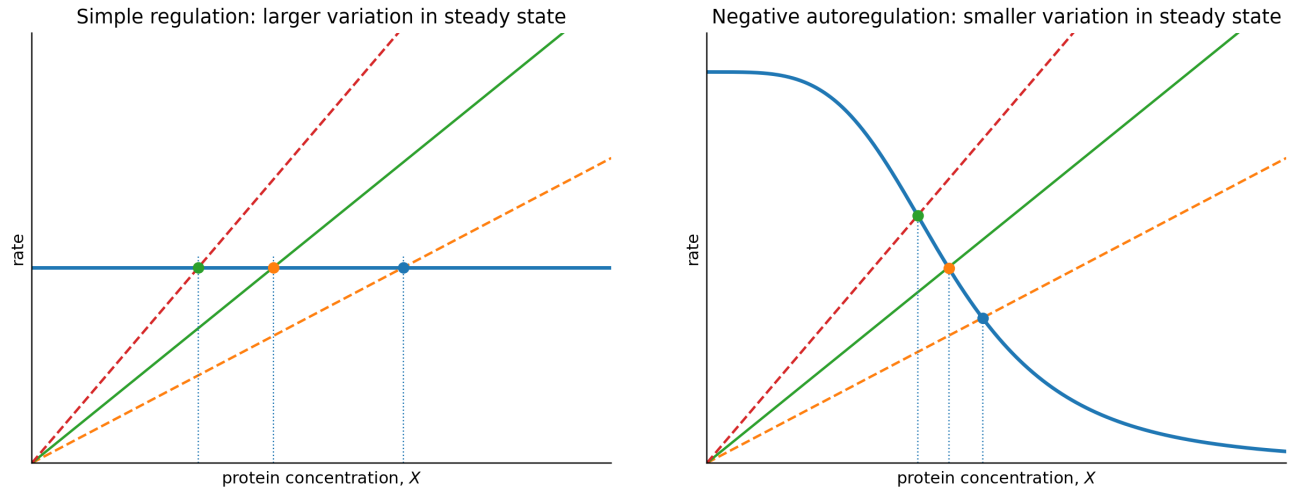


Figure 4. Left: simple regulation converts differences in removal rate into relatively large differences in steady-state protein concentration. Right: negative autoregulation compresses those differences. Conceptual plots redrawn from Alon, Figs. 2.10-2.11.

Thus the second effect is better described as

reduced cell-to-cell variability / increased robustness to parameter fluctuations

rather than elimination of all molecular noise.

9. Summary

After coarse-graining the fast regulatory and mRNA processes, the dynamics of a target protein concentration are described by

$$\frac{dy}{dt} = \beta f(x) - (\gamma + \mu)y.$$

The three terms have direct physical interpretations:

- $\beta f(x)$: effective protein production;
- γy : active protein degradation;
- μy : dilution caused by cell growth.

For a fixed input, the steady-state concentration is

$$y_{ss} = \frac{\beta f(x)}{\gamma + \mu}$$

and the half-response time is

$$T_{1/2} = \frac{\ln 2}{\gamma + \mu}.$$

Therefore, in the simplest gene-expression model:

- promoter strength controls **how much** protein accumulates at steady state;
- protein degradation and cell growth control **how fast** the system approaches that steady state;
- for a stable protein, dilution alone gives a response time of approximately one generation;
- negative autoregulation can make the response faster and can buffer cell-to-cell differences in kinetic parameters.

References

1. Alon, U. *An Introduction to Systems Biology: Design Principles of Biological Circuits*, 2nd ed. CRC Press. Chapters 1-2 and Appendix A.
2. Milo, R. & Phillips, R. *Cell Biology by the Numbers*. Garland Science.
3. Monod, J., Wyman, J. & Changeux, J.-P. On the nature of allosteric transitions: a plausible model. *J. Mol. Biol.* **12**, 88-118 (1965).
4. EcoCyc: experimentally curated information on *E. coli* genes and transcriptional regulation.